

Yateeka Goyal

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Research Interests

Intelligent and adaptive networking; quality of experience for video streaming; human-centered computing; behavioral data analysis; graph machine learning; cybersecurity and privacy.

Education

Georgia State University

M.S. in Computer Science, Concentration in Security and Privacy
GPA: 3.97/4.0

Atlanta, Georgia

Spring 2026 - Spring 2027

Relevant coursework: Machine Learning, Artificial Intelligence, Cryptography, Computer Networks, Advanced Computer Networks, Statistical Learning

Georgia State University, Honors College

B.S. in Computer Science; Undergraduate Certificate in Data Science
GPA: 3.91/4.0; Summa Cum Laude

Atlanta, Georgia

Fall 2022 - Fall 2025

Publication

M. E. Aktas, I. Moradi, I. Cosar, **Y. Goyal**, A. Alyaz, E. Akbas, and M. Ahsen, "Expert Detection on Crowdsourcing Forums via Sheaf Laplacian," in *2025 IEEE 12th International Conference on Data Science and Advanced Analytics (DSAA)*, 2025.

doi:10.1109/DSAA65442.2025.11247970.

Research Experience

Graduate Researcher, HESTIA Lab

Georgia State University

Atlanta, Georgia

2026 – Present

- Investigate how implicit behavioral signals can estimate when changes in video-streaming quality are most disruptive to viewers.
- Construct a dataset spanning sports, gaming, education, entertainment, podcast, and instructional videos using replay-intensity patterns, timestamped comments, content characteristics, and computational QoE metrics.
- Design a small-scale subjective user study to establish ground-truth viewer-sensitivity labels and assess relationships among behavioral signals and conventional QoE measures.
- Explore how viewer-sensitivity estimates can inform content-aware bitrate adaptation and network-resource allocation.

Undergraduate Researcher

Georgia State University

Atlanta, Georgia

2023 – 2025

- Contributed to a graph-based framework for detecting experts at the subtopic level in online crowdsourcing communities.
- Defined the initial problem scope, collected and preprocessed large-scale Stack Overflow data, and helped construct interaction-network representations.
- Identified relevant tags and interaction structures used in downstream sheaf Laplacian modeling and experimental evaluation.
- Collaborated with faculty and student researchers on research documentation, analysis, and preparation of the DSAA 2025 publication.

Teaching Experience

Graduate Teaching Assistant

Georgia State University, CSC 3210: Computer Organization and Programming

Atlanta, Georgia

Spring 2026

- Supported students across two course sections through office hours, assignment guidance, and explanations of computer organization and low-level programming concepts.
- Assisted the instructor with course administration, student questions, and assessment activities.

Undergraduate Teaching Assistant

Georgia State University, CSC 4222: Fundamentals of Cybersecurity

Atlanta, Georgia

Fall 2025

- Guided students through foundational cybersecurity concepts and supported course assignments and classroom activities.
- Assisted with grading, student questions, and instructional coordination.

Student Assistant and Tutor

Georgia State University TRIO Student Support Services

Atlanta, Georgia

Fall 2025

- Provided individualized academic assistance and helped students strengthen technical problem-solving and study skills.

Professional Experience

Graduate Administrative Assistant

Student Orientation & Family Engagement, Georgia State University

Atlanta, Georgia

Aug. 2026 – Present

- Develop a password-protected digital inventory-tracking system that uses barcode scanning to register, add, remove, and retrieve departmental items.
- Design a product-identification workflow that checks existing records, assigns new barcodes, and updates item quantities through a user-facing interface.
- Build an Excel-based data layer to limit hosting requirements while supporting inventory reporting and controlled access for registered users.
- Support family-engagement programs, outreach, recruitment, and university-wide events.

Data Science Intern

Student Orientation & Family Engagement, Georgia State University

Atlanta, Georgia

May 2026 – Aug. 2026

- Cleaned, transformed, and analyzed operational and service datasets using Python, NumPy, SQL, and Microsoft Excel.
- Automated call-log and weekly reporting workflows to identify recurring questions, service trends, and operational bottlenecks.
- Created summary metrics, dashboards, and technical reports to support departmental planning and data-informed decisions.

Selected Projects

CipherScope: LLM-Based Classification of Encryption Algorithms

Python, ML, LLMs

- Developed a full-stack application that predicts symmetric encryption algorithms from ciphertext using statistical features, machine-learning models, and large language models.
- Built automated preprocessing, feature-extraction, model-evaluation, and real-time inference workflows in Python.
- Deployed an interactive interface for ciphertext submission and algorithm prediction: cipherscope.onrender.com.

Call Log Analytics System

Python, Data Analysis, Excel

- Designed an analytics workflow to process support-call records, calculate performance metrics, and identify recurring service patterns.
- Automated recurring reports to improve consistency and reduce manual analysis effort.

Awards & Recognition

- First Place, Georgia State University Demo Day, Machine Learning Poster, 2024.
- University Assistantship Program Award, Georgia State University, 2025–2026.
- Summa Cum Laude, Georgia State University.
- Dean's List, four semesters; President's List, two semesters.

Presentations

- “Expert Detection on Crowdsourcing Forums via Sheaf Laplacian,” Machine Learning Poster, Georgia State University Demo Day, 2024.
- “Campus Connect,” Georgia State University Student Undergraduate Research Symposium, 2025.

Leadership & Service

Co-President

Girls Who Code College Loop at Georgia State University

Atlanta, Georgia

Jan. 2025 – Present

- Lead technical, professional-development, and community-building programs for students interested in computing.

Director

HackHERS 2026

Atlanta, Georgia

2025 – Present

- Lead a cross-functional team of more than 15 organizers in producing an MLH-approved hackathon for over 150 participants.
- Coordinate sponsor relations, programming, logistics, marketing, judges, mentors, speakers, and volunteer operations.
- Secured relationships with more than 10 corporate sponsors and community partners.

Technical Skills

Programming: Python, SQL, MySQL, TypeScript, JavaScript

Data Science and Machine Learning: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis, Model Evaluation, Experimental Design

Research Areas: Graph Machine Learning, Network Analysis, Quality of Experience, Behavioral Data Analysis, Cybersecurity

Tools: Git, GitHub, Linux, Jupyter Notebook, Microsoft Excel, LaTeX